Optimal Control HW6

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ME EN 7210 Optimal Controls **Spring 2026**
Assignment #6 **Due Thursday, March 26 at the beginning of class**

This assignment MUST be typeset in either LaTex or MS Word. Any hand-written work will not be accepted and if submitted, the entire assignment will receive a zero grade! Please start on this assignment early and come see Dr. Leang with any questions.

Problem 1 Find the optimal feedback control for the plant

$$\dot{x}_1(t) = x_2(t)$$
$$\dot{x}_2(t) = -2x_1(t) + 4x_2(t) + 5u(t)$$

with performance criterion

$$J = \frac{1}{2} f_{11} \left[x_1(t_f) - 4\right]^2 + \frac{1}{2} f_{22} \left[x_2(t_f) - 2\right]^2 + \frac{1}{2} \int_0^{t_f} \left[5x_1^2(t) + 2x_2^2(t) + 4u^2(t)\right] dt$$

and initial conditions as $\mathbf{x}(0) = [1 \quad 2]'$ and the final state $\mathbf{x}(t_f)$ is free, where t_f is specified.

Assume that the final time is 5. Be sure to plot the Riccati coefficients, optimal states, and optimal input as a function of time.

Problem 2: Optimal Control of a Motor System

Consider the Quanser motor system shown in Fig. 1 below (System from the Small Robotics Lab). Note, this system should be familiar to you from other courses, such as ME EN 5200/6200 or 5210/6210.

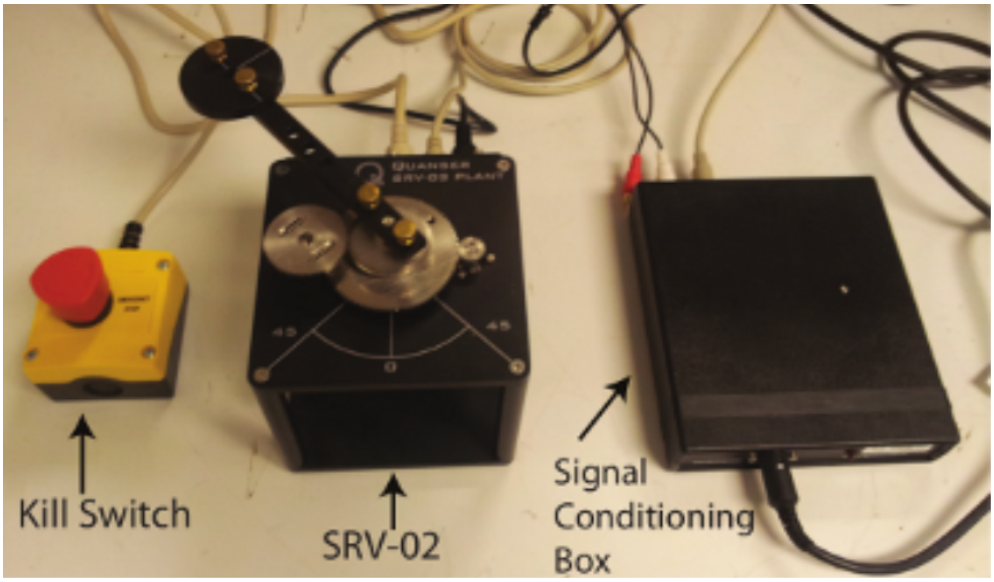


Figure 1. The system has a motor/mass system (SRV-02) connected to a computer via an interface board (NI-6024).

The system is typically configured as a position feedback control system and can be modeled by the following block diagram:

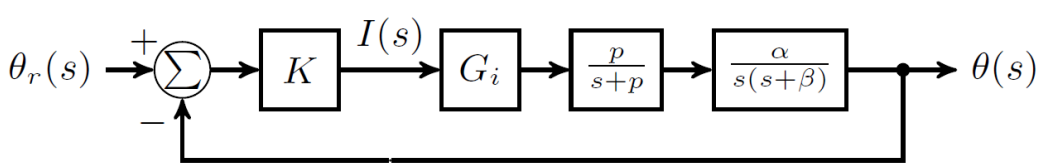


Figure 2. Closed-loop system block diagram.

In Fig. 2, $G_i = 10$ is the drive gain, $p = 20$ is the filter parameter, $\alpha = 7{:}5$ and $\beta = 1{:}67$ are constants found experimentally, I is the commanded current input, θ is the rotational position, and θ_r is the desired (or reference) rotational position. Note that the transfer function $\alpha/(s(s+\beta)$ is the motor transfer function. Note that the open-loop model contains G_i, the filter, and the transfer function of the motor.

(a) Before doing any design work on the controller, your first task is look at the dynamics of the motor system in simulation. Do to this, you will need to apply a series of inputs I(t) and simulate the response, then plot the simulated response using Matlab. Show the details of this step, along with plots to justify the overall behavior of the open-loop system model. Explain what you see and whether the simulated response makes sense in terms of how it models the real physical system.

(b) Design a finite-time full-state LQR controller for the motor-mass system. It should minimize the energy (e.g., control effort and energy in the states). It should start from rest at angle 0° and move to 45°. Let the final time be 1 second.

Define the weighting matrices, F, Q, and R.
Try the following cases:

$Q = 0$ and $F \neq 0$
$F = 0$ and $Q \neq 0$
$F \neq 0$ and $Q \neq 0$

Plot the results, including the gains vs. time, response vs. time, and input vs. time.

Problem 1

$$H = \frac{5}{2}x_1^2 + x_2^2 + 2u + \lambda_1 x_2 + \lambda_2(-2x_1 + 4x_2 + 5u)$$

$$\frac{\partial H}{\partial u} = 4u + 5\lambda_2 = 0$$

$$u^* = -\frac{5}{4}\lambda_2$$

$$H^* = \frac{5}{2}x_1^2 + x_2^2 + \frac{25}{8}\lambda_2^2 + \lambda_1 x_1 + \lambda_2(-2x_1 + 4x_2 - \frac{25}{4}\lambda_2)$$

$$\dot{x}_1^* = \frac{\partial H^*}{\partial \lambda_1} = x_2$$

$$\dot{x}_1^* = \frac{\partial H^*}{\partial \lambda_2} = 4x_2 - 2x_1 - \frac{25}{4}\lambda_2$$

$$\dot{\lambda}_1^* = -\frac{\partial H^*}{\partial x_1} = -5x_1 + 2\lambda_2$$

$$\dot{\lambda}_2^* = -\frac{\partial H^*}{\partial x_2} = -2x_2 - \lambda_1 - 4\lambda_2$$

$$\lambda_1^*(t_f) = \frac{\partial S}{\partial x_1(t_f)} = f_{11}(x_1 - 4)$$

$$\lambda_2^*(t_f) = \frac{\partial S}{\partial x_2(t_f)} = f_{22}(x_1 - 2)$$

$$\lambda^*(t) = p(t)x(t) - g(t)$$

$$\begin{bmatrix} \lambda_1 \\ \lambda_2 \end{bmatrix} = \begin{bmatrix} P_{11} & P_{12} \\ P_{21} & P_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} - \begin{bmatrix} g_1 \\ g_2 \end{bmatrix}$$

$$\lambda_1^* = P_{11}x_1 + P_{12}x_2 - g_1$$

$$\lambda_2^* = P_{21}x_1 + P_{22}x_2 - g_2$$

$$\dot{\lambda}_1^* = \dot{P}_{11}x_1 + P_{11}\dot{x_1} + \dot{P}_{12}x_2 + P_{12}\dot{x}_2 - \dot{g}_1$$

$$\dot{\lambda}_2^* = \dot{P}_{21}x_1 + P_{21}\dot{x_1} + \dot{P}_{22}x_2 + P_{22}\dot{x}_2 - \dot{g}_2$$

Equate co-state derivative results from the Hamiltonian with the generic costate derivative found above with ricatti coefficients:

$$-5x_1 + 2\lambda_2 = \dot{P}_{11}x_1 + P_{11}\dot{x_1} + \dot{P}_{12}x_2 + P_{12}\dot{x}_2 - \dot{g}_1$$

$$-2x_2 - \lambda_1 - 4\lambda_2 = \dot{P}_{21}x_1 + P_{21}\dot{x_1} + \dot{P}_{22}x_2 + P_{22}\dot{x}_2 - \dot{g}_2$$

Subbing in expressions for $\dot{x}_1, \lambda_2, \dot{x}_2, \lambda_1$ results in the following differential equations for the co-states:

$$\dot{P}_{11} = 4P_{12} + \frac{25}{4}P_{12}^2 - 5$$

$$\dot{P}_{12} = 2P_{22} - P_{11} - 4P_{22} + \frac{25}{4}P_{12}P_{22}$$

$$\dot{P}_{22} = -2P_{12} - 8P_{22} + \frac{25}{4}P_{22}^2 - 2$$

$$\dot{g}_1 = \frac{25}{4}P_{12}g_2 + 2g_2$$

$$\dot{g}_2 = \frac{25}{4}P_{22}g_2 - 4g_2 - g_1$$

These differential equations need to be solved backward in time with the following final conditions:

$$P(t_f) = F(t_f) = \begin{bmatrix} f_{11} & 0 \\ 0 & f_{22} \end{bmatrix}, \; g(t_f) = \begin{bmatrix} 4f_{11} \\ 2f_{22} \end{bmatrix}$$

All Code is given in the Appendix

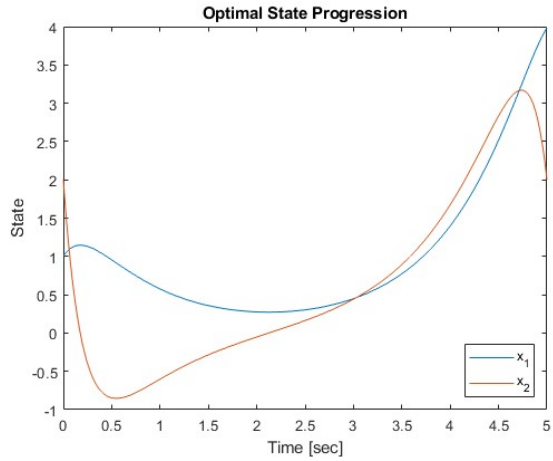


Figure 1: Optimal States vs Time

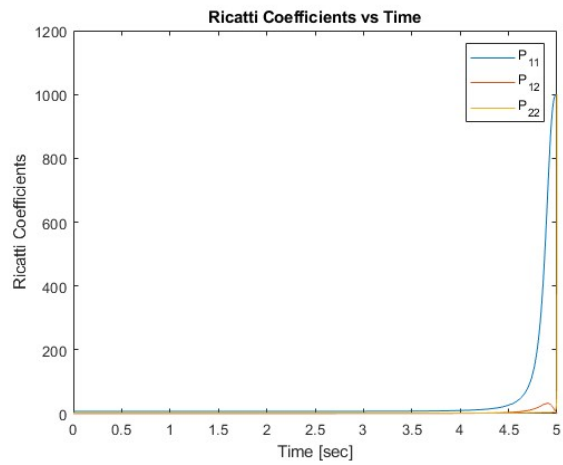


Figure 2: Ricatti Coefficients vs Time

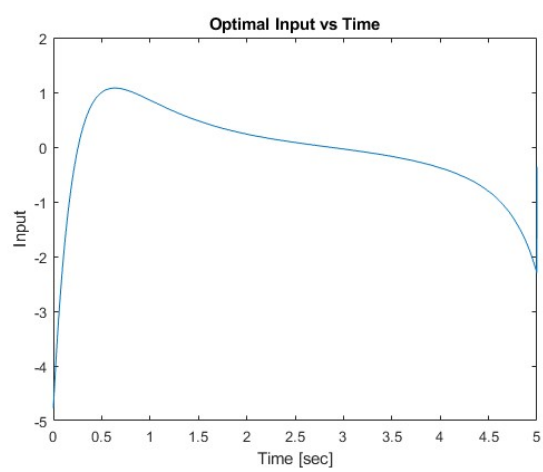


Figure 3: Optimal Input vs Time

Problem 2

Part a)
To understand the dynamics of the open-loop, 3 inputs were simulated using matlab, step input, ramp input, and parabolic input. This was done by obtaining the open loop transfer function given by:

$$\frac{\theta(s)}{I(s)} = \frac{G_i P \alpha}{(s+p)s(s+\beta)} = \frac{1500}{s^3 + 21.67s^2 + 33.4s}$$

The simulation used Matlabs Lsim function and the results are as shown:

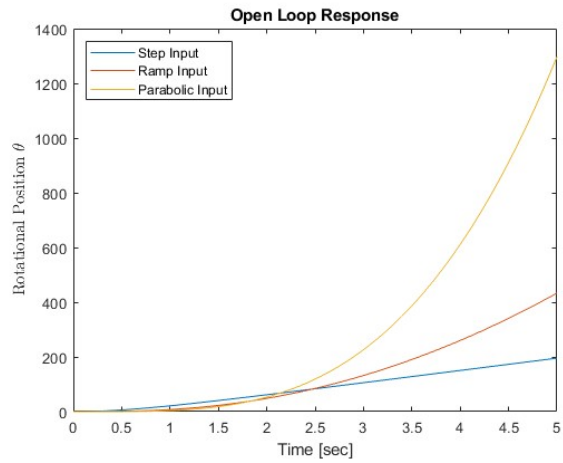


Figure 4: Open Loop Response to Different Inputs

These results for different inputs make sense according to how we would expect the real model to respond. For each input of current, the position continues to grow at a rate decided by the type of input. For the parabolic input, its most obvious that the position grows very fast with a parabolic trajectory. A step input results in linear growth in position and the ramp input results in a trajectory that slowly increases in time just like we would expect.

Part b)
To design an LQR controller for this system, it must be put into state equation form by doing the following:

$\frac{\theta(s)}{I(s)} = \frac{1500}{s^3+21.67s^2+33.4s}$

$\mathcal{L}^{-1}[\theta(s)(s^3 + 21.67s^2 + 33.4s)] = \mathcal{L}^{-1}[1500I(s)]$

$\dddot{\theta} + 21.67\ddot{\theta} + 33.4\dot{\theta} = 1500I$

$\dddot{\theta} = -21.67\ddot{\theta} - 33.4\dot{\theta} + 1500I$

Let $\dot{x}_1 = \dddot{\theta} \rightarrow x_1 = \ddot{\theta}$

Let $x_2 = \dot{\theta} \rightarrow \dot{x}_2 = \ddot{\theta} = x_1$

Let $x_3 = \theta \rightarrow \dot{x}_3 = \dot{\theta} = x_2$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -21.67 & -33.4 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 1500 \\ 0 \\ 0 \end{bmatrix} I$$

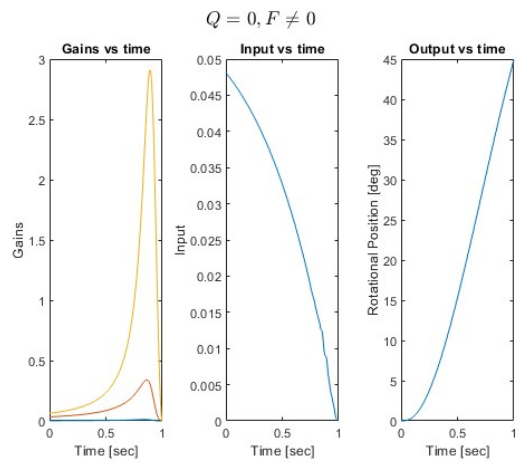
The cost function is given by:

$$J = \frac{1}{2}x(t_f)^T F(t_f)x(t_f) + \frac{1}{2}\int_{t_o}^{t_f}[x(t)^T u(t)^T]\begin{bmatrix} Q(t) & 0 \\ 0 & R(t) \end{bmatrix}\begin{bmatrix} x(t) \\ u(t) \end{bmatrix}$$

where $x(t_f) = \begin{bmatrix} 0 \\ 0 \\ \frac{\pi}{4} \end{bmatrix}$ and $x(0) = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

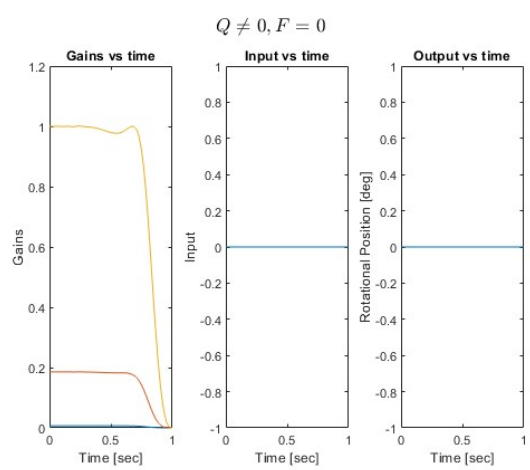
R was set to 1 to penalize the input and 3 cases of weightings were used:

Case 1: $Q = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, F = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$



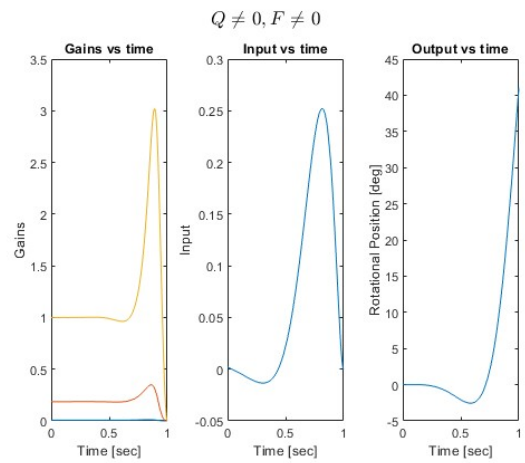
Here it can be seen that by not penalizing the state progression but penalizing the final state, the final state has met with relative ease.

Case 2: $Q = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, F = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$



Here it can be seen that by penalizing the state progression but not penalizing the final state, the output is kept regulated at zero with ease without meeting the final state condition because there is no importance placed on it.

Case 3: $Q = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, F = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$



Here it can be seen that by penalizing the state progression and the final state equally, the output has a harder time actually reaching the final desired state because the weighting on the state progression is holding it back from doing so. As a result, the final state is not met.

Appendix

```
% Problem 1
clear, clc, close all

%Initial Conditions
x0 = [1;2];

%Final State Weights
f11 = 1000;
f22 = 1000;

%Reversed time vector for backwards solving
t_rev = linspace(5,0,5000);

%Final Conditions
final_cond = [f11; 0; f22; 4*f11; 2*f22];

%Solving for P backwards in time and then flipping the solution
[t,z] = ode45(@odefun, t_rev ,final_cond);
t_up = flip(t);
z_up = flip(z,1);

%Solving for the optimal states from P
[t_x, x] = ode45(@(t,x) dynamics(t,x,t_up,z_up), flip(t_rev), x0);

%Returning the optimal input from the states and P
for i = 1:length(t_x) ...

%Optimal Dynamics
function dxdt = dynamics(t,x,t_up,z_up) ...

%Optimal Gains
function odes = odefun(t,z) ...

%Plotting
figure
plot(t_x,x)
title("Optimal State Progression"); xlabel("Time [sec]"); ylabel("State"); legend("x_1", "x_2", "Location","southeast")
figure
plot(t_up, z_up(:,1:3))
title("Ricatti Coefficients vs Time"); xlabel("Time [sec]"); ylabel("Ricatti Coefficients"); legend("P_{11}", "P_{12}", "P_{22}", "Location","north
figure
plot(t_x,u)
title("Optimal Input vs Time"); xlabel("Time [sec]"); ylabel("Input");
```

```
%%
clear, clc, close all

sys = tf([1500], [1 21.67 33.4 0]);

t = linspace(0,5,1000);
ramp = t;
parabola = t.^2;

[y_step,tOut] = step(sys,t);
[y_ramp,tOut] = lsim(sys,ramp,t);
[y_para,tOut] = lsim(sys,parabola,t);

figure
plot(tOut,y_step)
hold on
plot(tOut,y_ramp)
hold on
plot(tOut,y_para)
title("Open Loop Response"); xlabel("Time [sec]"); ylabel("Rotational Position $\theta$","Interpreter","latex"); legend("Step Input", "Ramp Input", "Parabolic Input", "Location", "

tf2ss([1500], [1 21.67 33.4 0])
```

```
%% Part B
clear,clc,close all
%State Matrix Equation
A = [-21.67 -33.4 0;
      1      0    0;
      0      1    0];

B = [1500; 0; 0];
C = eye(3);
%Time
tf = 1;
t = linspace(0,tf,1000);
%Weightings
R = 1;
Q = zeros(3);
F = [0 0 0;
    0 0 0;
    0 0 1];
%Initial and Final Conditions for States
x0 = [0;0;0];
xf = [0;0;pi/4];
%Preliminary Calculations
E = B*(1/R)*B';
V = C'*Q*C;
W = C'*Q;
%Solving For Riccatti Coefficients
[sol_P] = ode45(@(t,p) pFun(t, p, A, E, V), [tf,0], F(:));
P = deval(sol_P,t);
%Solving for final state contribution
sol_g = ode45(@(t,g) gFun(t,g,sol_P,A,E), [tf 0], F*xf);
G = deval(sol_g,t);
%Solving for states
sol_x = ode45(@(t,x) xFun(t,x,sol_P,sol_g,A,E), [0 tf], x0);
X = deval(sol_x,t);
%Solving for gains and optimal input
for i = 1:length(t)
    Pi = reshape(P(:,i),3,3);
    gi = G(:,i);
    xi = X(:,i);
    Ki = (1/R)*B'*Pi;
    K(:,i) = Ki(:);
    u(i) = -Ki*xi + (1/R)*B'*gi;
end
%Plotting
figure
subplot(1,3,1)
plot(t,K')
title("Gains vs time"); xlabel("Time [sec]"); ylabel("Gains");
subplot(1,3,2)
plot(t,u')
title("Input vs time"); xlabel("Time [sec]"); ylabel("Input");
subplot(1,3,3)
plot(t,X(3,:).*(180/pi))
title("Output vs time"); xlabel("Time [sec]"); ylabel("Rotational Position [deg]");
sgtitle("$Q = 0, F \neq 0$", 'interpreter', 'latex')
```